

NORRKOPING KUNGSANGEN, Sweden

WMO: 025710

Lat: 58.586N

Lon: 16.251E

Elev: 10

StdP: 101.21

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-16.1	-12.9	-18.1	0.8	-15.7	-15.1	1.0	-10.7	12.8	5.4	11.2	4.9	1.5	260	0.628

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.3	27.1	17.8	25.2	17.3	23.8	16.6	19.6	24.3	18.5	23.1	17.5	21.9	4.1	260

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.1	13.0	20.9	16.9	12.1	20.1	15.9	11.3	19.3	55.7	24.4	52.3	22.8	49.4	21.7	23.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 10.5	9.2	8.1	DB	-18.5	29.8	4.7	2.1	-21.8	31.4	-24.6	32.6	-27.2	33.8	-30.6	35.3	(4)
(5)			WB	-18.7	21.0	4.7	1.3	-22.1	21.9	-24.9	22.7	-27.5	23.4	-30.9	24.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	7.3	-2.1	-1.9	1.3	6.1	11.0	14.8	17.3	16.2	12.5	7.4	3.5	0.3	(6)
(7)		DBStd	7.73	4.96	5.05	4.17	3.37	3.44	2.75	2.62	2.53	2.86	3.48	3.55	4.80	(7)
(8)		HDD10.0	1731	374	332	271	122	29	2	0	1	10	93	195	301	(8)
(9)		HDD18.3	4072	633	566	529	366	227	109	51	74	176	338	444	559	(9)
(10)		CDD10.0	734	0	0	0	7	61	146	227	193	85	14	1	0	(10)
(11)		CDD18.3	34	0	0	0	0	0	4	21	9	1	0	0	0	(11)
(12)		CDH23.3	423	0	0	0	1	19	83	227	89	4	0	0	0	(12)
(13)		CDH26.7	74	0	0	0	0	1	12	47	14	0	0	0	0	(13)
(14)	Wind	WSAvg	3.7	3.8	3.9	4.0	3.7	3.5	3.7	3.4	3.3	3.5	3.6	3.8	4.0	(14)
(15)	Precipitation	PrecAvg	609	38	36	28	35	42	65	75	72	58	53	57	48	(15)
(16)		PrecMax	785	72	63	54	84	79	110	139	162	137	139	124	93	(16)
(17)		PrecMin	507	5	14	7	3	9	11	4	9	9	16	11	12	(17)
(18)		PrecStd	60	18	12	13	23	18	25	34	36	36	29	28	23	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	8.1	9.2	14.8	20.2	25.2	28.2	30.1	28.2	24.0	17.2	12.8	11.8	(19)
(20)			MCWB	6.9	6.9	8.3	11.9	15.6	17.2	18.7	18.0	16.3	14.0	10.8	9.4	(20)
(21)		2%	DB	6.9	7.1	11.9	17.8	23.1	25.9	28.0	25.9	21.8	15.1	11.0	9.1	(21)
(22)			MCWB	5.7	5.2	7.0	10.6	14.4	17.0	18.4	17.5	15.6	12.8	9.2	7.7	(22)
(23)		5%	DB	5.2	5.9	9.2	15.2	20.9	23.8	26.0	23.8	19.2	14.0	9.9	7.8	(23)
(24)			MCWB	4.1	4.4	5.6	9.5	13.7	16.1	17.9	16.9	14.7	12.0	8.6	6.5	(24)
(25)		10%	DB	4.1	4.2	7.2	13.1	18.2	21.4	24.0	22.0	18.0	12.9	8.8	6.1	(25)
(26)			MCWB	3.0	2.9	4.6	8.0	12.2	14.8	17.3	16.2	13.9	11.0	7.9	5.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.1	7.2	9.1	13.4	17.5	19.4	21.4	20.8	17.7	14.7	11.2	9.6	(27)
(28)			MCDWB	7.7	8.6	13.7	18.4	22.4	24.4	25.2	24.5	21.1	16.7	12.6	11.0	(28)
(29)		2%	WB	5.5	5.5	7.2	11.2	15.7	17.8	20.0	19.1	16.3	13.4	9.6	7.8	(29)
(30)			MCDWB	6.3	6.6	10.4	16.0	20.9	23.3	25.0	22.9	19.9	15.0	10.5	8.9	(30)
(31)		5%	WB	4.1	4.2	6.0	10.0	14.3	16.7	19.0	18.0	15.4	12.4	8.6	6.3	(31)
(32)			MCDWB	5.0	5.3	8.4	14.8	19.1	22.0	23.9	21.5	18.6	13.9	9.5	7.1	(32)
(33)		10%	WB	2.9	3.0	4.8	8.5	13.0	15.6	18.0	17.0	14.6	11.3	7.6	5.0	(33)
(34)			MCDWB	3.7	4.1	6.9	12.2	17.6	20.4	22.9	20.6	17.3	12.8	8.3	5.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.1	6.0	7.8	10.5	11.0	10.5	10.3	10.2	9.2	7.1	4.9	4.9	(35)
(36)			MCDBR	5.4	6.0	10.7	14.8	15.9	15.3	14.4	13.5	11.8	8.2	6.0	5.3	(36)
(37)			MCWBR	5.0	4.7	7.0	8.9	8.4	7.6	6.4	6.7	7.1	6.2	5.2	4.9	(37)
(38)		5% WB	MCDBR	5.2	5.4	9.6	13.2	13.1	12.8	12.2	11.0	10.5	7.5	5.7	5.3	(38)
(39)			MCWBR	5.0	4.5	6.9	8.4	7.7	7.0	6.2	6.3	6.8	6.0	5.2	5.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.286	0.323	0.342	0.380	0.376	0.364	0.392	0.383	0.350	0.340	0.312	0.273	(40)	
(41)		taud	2.242	2.293	2.346	2.303	2.382	2.439	2.383	2.405	2.481	2.414	2.288	2.178	(41)	
(42)		Ebn at Noon	542	697	800	824	856	872	835	815	790	676	502	421	(42)	
(43)		Edh at Noon	46	71	89	109	108	104	107	98	78	63	44	35	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.34	0.93	2.43	3.98	5.26	5.81	5.29	4.21	2.82	1.26	0.43	0.23	(44)	
(45)		RadStd	0.05	0.16	0.25	0.40	0.43	0.27	0.51	0.40	0.34	0.20	0.09	0.03	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (55 neighbors)	+0.39	N/A	N/A	N/A	N/A	N/A	-129	-158	N/A	N/A

Nomenclature: See separate page